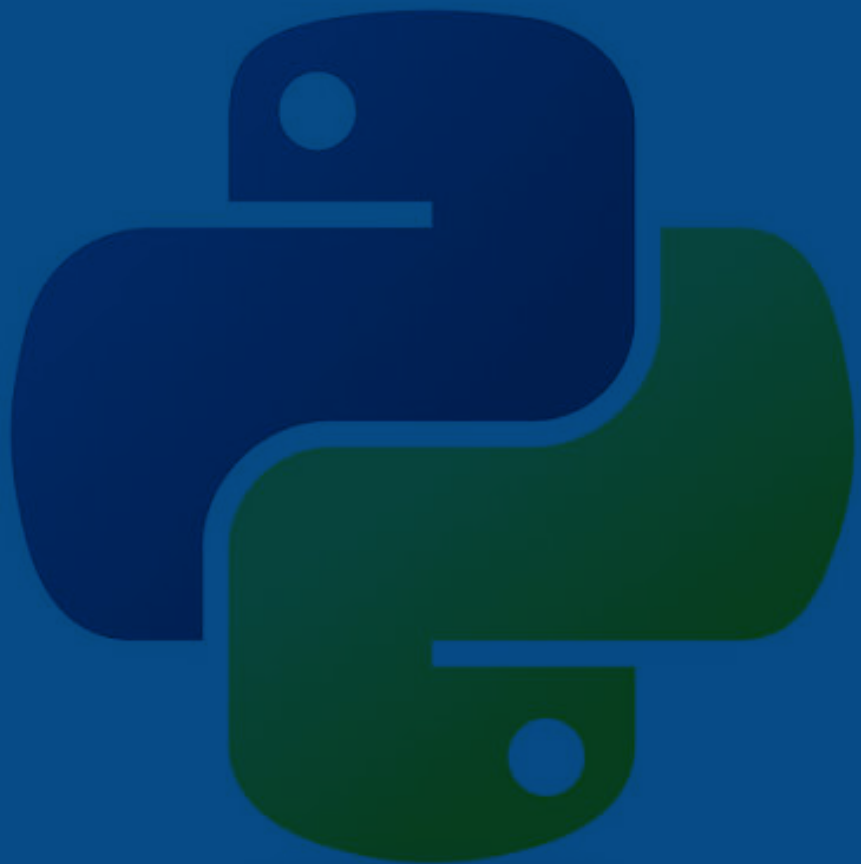


Python Installation and Setup

GETTING STARTED
WITH PYTHON FOR
INDUSTRIAL AND
INFORMATION
ENGINEERING

A STEP-BY-STEP GUIDE TO INSTALLING PYTHON AND
SETTING UP THE DEVELOPMENT ENVIRONMENT



CONTENUTI

- | | |
|-----------------------|------|
| 1. INSTALLING PYTHON | p. 2 |
| 2. INSTALLING VS CODE | p. 5 |

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Contents

1	Installing Python	2
1.1	Windows	3
1.1.1	Option 1: app store install	3
1.1.2	Option 2: direct install	3
1.2	MacOS	4
1.2.1	Option 1: graphical install	4
1.2.2	Option 2: using Homebrew	4
1.3	Linux	4
1.4	Mobile Devices	5
2	Installing VS Code	5

SECTION 1

Installing Python

1.1 WINDOWS

1.1.1 OPTION 1: APP STORE INSTALL

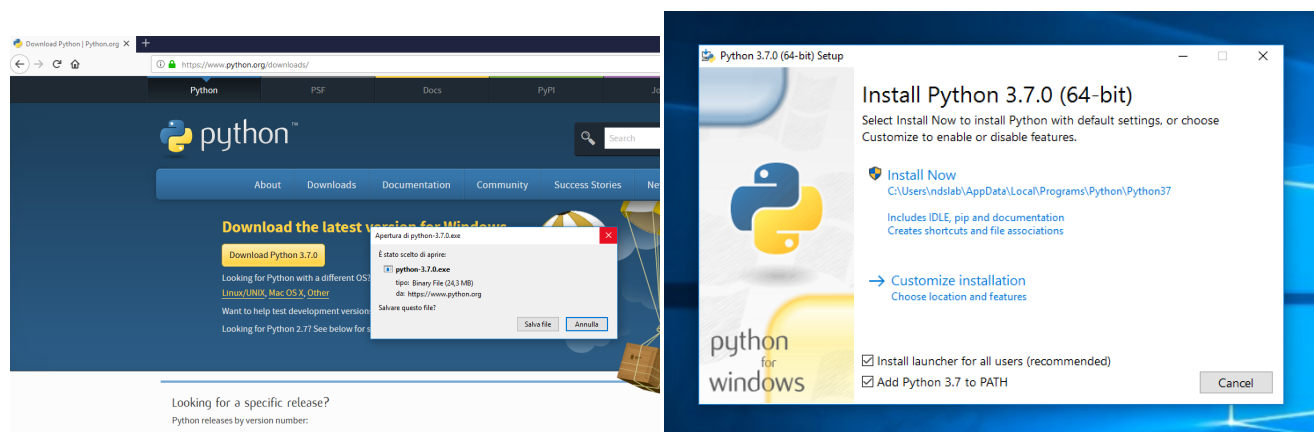
The easiest way to install Python on Windows is using the Microsoft Store: <https://apps.microsoft.com/search?query=python>

1. Open the Start menu (Windows icon in the bottom left), type “Microsoft Store”, and select the link to open the store.
2. Once the store is open, select Search from the top-right menu and enter “Python”. Select Python 3.12 from the results under Apps, then select Get.
3. Once the download and installation process is complete, open Windows PowerShell from the Start menu (Windows icon in the bottom left).
4. Once PowerShell is open, enter `python -version` to confirm that Python was installed correctly.
5. The Python installation includes pip. Pip allows you to install and manage additional packages. To confirm that pip is also available to install and manage packages, type `pip -version`.

1.1.2 OPTION 2: DIRECT INSTALL

Go to the official Python website and download the latest version of Python for Windows: <https://www.python.org/downloads/>.

After downloading the installer, run it and follow the installation prompts. Make sure to check the box that says “Add Python 3.12 to PATH” before clicking “Install Now”.



To verify that Python was installed correctly, open Windows PowerShell and type `python -version`.

Manually adding Python to PATH

If you did not check the box to add Python to PATH during installation, or if something went mysteriously wrong, you can manually add Python to your PATH environment variable.

There are many guides online that explain how to do this, but here is a brief summary:

1. Open the Start menu and search for “Environment Variables”.
2. Click on “Edit the system environment variables”.
3. In the System Properties window, click on the “Environment Variables” button.
4. In the Environment Variables window, under “System variables”, find the variable named “Path” and select it, then click “Edit”.
5. In the Edit Environment Variable window, click “New” and add the path to the Python installation directory (It can be found by searching for “Python” in the Start menu, right-clicking on the Python app, and selecting “Open file location”).
6. Click “OK” to close all windows.
7. Open a new PowerShell window and type `python --version` to verify that Python is now recognized.

1.2 MACOS

1.2.1 OPTION 1: GRAPHICAL INSTALL

1. Go to the official Python website: <https://www.python.org/downloads/>
2. Click on the “Download Python” button for the latest version.
3. Open the downloaded .pkg file.
4. Follow the installer prompts (Continue > Agree > Install) and enter your Mac password when requested.
5. Once the installation is complete, you can verify the installation by opening a terminal (Cmd + Space and typing “Terminal”) and typing:
`python3 --version`

1.2.2 OPTION 2: USING HOMEBREW

If you prefer using the command line, you can install Python using Homebrew, a popular package manager for macOS. If you don’t have Homebrew installed, you can install it by following the instructions on the official website: <https://brew.sh/>.

Once Homebrew is installed, you can install Python by opening a terminal and typing:

```
brew install python
```

1.3 LINUX

Good news! Python is already installed on most Linux distributions. You can check if Python is installed by opening a terminal and typing:

```
python3 --version
```

If Python is not installed, you can install it using your package manager. For example, on Ubuntu, you can use the following command:

```
sudo apt update
sudo apt install python3
```

1.4 MOBILE DEVICES

While it is possible to run Python on mobile devices, it is **not** recommended for this course. Writing and running Python code on a mobile device can be cumbersome and may not provide the best learning experience.

Nevertheless, if you cannot access a computer in class, even a suboptimal experience is better than no experience at all.

In past course editions, students have used the following apps to run Python on mobile devices with some success (your mileage may vary):

- **Android:**
 - Pydroid 3
 - QPython
- **iOS:**
 - Carnets
 - Pyto
 - Pythonista (paid app)
- **Browser-based:**
 - <https://trinket.io/>
 - <https://repl.it/>
 - <https://www.online-python.com/>
 - <https://colab.research.google.com/>

Ergonomics warning:

Using a mobile device for coding can be uncomfortable and may lead to strain or injury over time. If you must use a mobile device, it is highly recommended to use an external bluetooth keyboard and a stand to position the device at a comfortable viewing angle.

A tablet is preferable to a phone, as it provides a larger screen, reducing eye strain and making it easier to read long lines of code.

SECTION 2

Installing VS Code

Installing Visual Studio Code (VS Code) is not strictly necessary, but it is highly recommended for a better coding experience. VS Code is a free and open-source code editor developed by Microsoft, which supports many programming languages, including Python.

VS Code can be downloaded from the official website, <https://code.visualstudio.com/Download>, by selecting the appropriate version for your operating system (Windows, MacOS, or Linux).

For a better experience, it is recommended to install the Python extension for VS Code: <https://marketplace.visualstudio.com/items?itemName=ms-python.python>.

Please follow this excellent official guide for more information on how to use the Python extension in VS Code: <https://code.visualstudio.com/docs/languages/python>

Do **not** enable AI features in VS Code, as they are antithetical to the learning objectives of this course.